

Jayant Prakash

Forward Deployed Engineer

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Summary

Forward Deployed Engineer/AI Engineer with 8+ years of experience designing, building, and deploying production AI/ML systems, LLM applications, RAG pipelines, and scalable GenAI platforms. Strong expertise in OpenAI models, Azure AI Foundry, agent workflows, vector search, AI evaluation, hallucination detection, factual accuracy, guardrails, Responsible AI, and ML model development. Experienced in API development, system design, cloud-native architecture, FastAPI, Docker, monitoring, and production deployment across AWS and Azure.

Professional Experience

Forward Deployed Engineer | AI Engineer | (2025-2026)

AT&T

- Architected, developed, and deployed an agentic OpenAI/LLM-powered outage intelligence system to process noisy and unstructured incident logs with inconsistent formats and missing fields, reducing manual triage by ~70% and improving response time by ~45%, driving operational efficiency and cost savings.
- Partnered directly with business, operations, and engineering stakeholders to identify high-value automation opportunities, translate requirements into solution designs, and integrate AI services with enterprise platforms.
- Developed REST API integrations to retrieve operational and incident data for AI agents and updated ServiceNow tickets through REST APIs, enabling automated incident enrichment, orchestration, and response management.
- Architected modular agent workflows using LangGraph, OpenAI models, tool-calling, retrieval, reasoning, and validation for telecom use cases.
- Engineered prompt pipelines with structured outputs, guardrails, factuality checks, and failure handling to reduce hallucination, achieving ~90% classification accuracy.
- Developed RAG pipelines using chunking, embeddings, hybrid retrieval, re-ranking and reranker methods, Pinecone, vector search, and context enrichment to improve grounding, contextual relevance, and response accuracy.
- Developed responsive, enterprise-grade frontend applications at AT&T using HTML5, CSS3, JavaScript, and React.js, building reusable UI components, integrating REST APIs, and delivering scalable user experiences aligned with business and operational requirements.
- Implemented AI evaluation and testing framework (unit, integration, prompt regression, hallucination detection, factual accuracy, bias checks, LLM-as-judge), reducing production failures by ~60%.
- Developed and deployed FastAPI-based APIs, containerized using Docker, enabling production rollout with monitoring, logging, and reliability validation under real-world usage scenarios.
- Mentored junior engineers on GenAI development, RAG pipelines, prompt engineering, and AI evaluation frameworks across telecom AI initiatives.

AI Engineer | ML Engineer (2021-2025)

Comcast Corporation

- Designed, developed, and deployed an Azure AI Foundry-enabled Generative AI dispute resolution system, reducing processing time by ~70% through automation.
- Built LLM pipelines using Llama 70B and Azure AI Foundry to process customer communications, enabling automation of ~35% of disputes.
- Fine-tuned LLM models using Azure AI Foundry to improve dispute classification accuracy, grounding, and response consistency across customer communication workflows.
- Engineered recommendation system improving acceptance from 23% to 95%, reducing manual allocation effort.
- Classified social media sentiment using embeddings and neural networks in TensorFlow to improve customer insights. Tuned hyperparameters using Keras Tuner, achieving test accuracy of 98%.
- Developed and consumed RESTful APIs to ingest, preprocess, and deliver data pipelines that support machine learning workflows.
- Wrote SQL queries in Teradata to collect, validate, and prepare data for machine learning models.

- Used GitHub Actions for CI/CD and MLOps, with Prometheus and Grafana for monitoring, visualization, and reliability tracking.
- Mentored junior engineers and cross-functional teams on LLM application development and ML best practices.

ML Engineer (2019-2020)

Pathology Dynamics Lab | Georgia Institute of Technology

- Predicted Alzheimer's cognitive decline given BACE inhibitor drugs and other features using XGBoost, achieving R^2 score of 0.85.
- Segmented Alzheimer's patients into four clusters using PCA and k-means in Python to support personalized treatment. Deployed the ML model to production using AWS SageMaker.

ML Engineer (2015-2018)

American Airlines | TCS

- Performed market segmentation using k-means in Python on airline customer data to increase the retention of customers by 10% using targeted marketing campaigns.

Full Stack Software Engineer (2012-2015)

SABMiller | TCS

- Developed enterprise SAP front-end applications in Java supporting purchase order workflows.

Education

Master of Science in Computer Science, Concentration in Machine Learning (2020)

Georgia Institute of Technology, Atlanta, Georgia

Bachelor of Technology in Computer Science (2012)

SRM University, Chennai, India

Technical Proficiencies

Programming & Core Engineering: Python, Java, Data Structures & Algorithms, System Design, OOPS

Large Language Models and AI Agents: OpenAI models, Azure AI Foundry, LangGraph, LangChain, Prompt Engineering, RAG, AI Agents, Tool Calling, Guardrails, Hallucination Detection, Factual Accuracy, Responsible AI, Fine-tuning, Vector Search, Pinecone, FAISS, ChatGPT, Llama 3, Claude, Gemini

Machine Learning & Deep Learning: XGBoost, Random Forest, SVM, neural networks, CNN, RNN, LSTM, Transformers, NLP, Computer Vision, recommendation systems, time series forecasting, PCA, t-SNE, K-Means, DBSCAN

Data & Libraries: PyTorch, TensorFlow, Scikit-learn, Keras, Pandas, NumPy, Matplotlib

MLOps & Production Systems: FastAPI, REST APIs, Docker, AWS SageMaker, AWS EC2, AWS S3, Azure, Azure AI Foundry, GitHub Actions, Prometheus, Grafana, MLflow, SQL Server, Teradata

Research Papers

- Quantifying amyloid beta reduction necessary in transgenic Alzheimer's mice using BACE inhibitors. Journal of Alzheimer's Disease. – ([View Paper](#))
- Unsupervised Machine Learning to Identify Separable Clinical Alzheimer's Disease Sub-Populations – ([View Paper](#))
- Biomedical Text Link Prediction using Graph Neural Network for Drug Discovery: A Case Study with COVID-19 – ([View Paper](#))

Certifications

- AI Agents – ([View Certificate](#))
- Machine Learning – Issued by Stanford University – ([View Certificate](#))
- Inferential Statistics – Issued by Duke University – ([View Certificate](#))